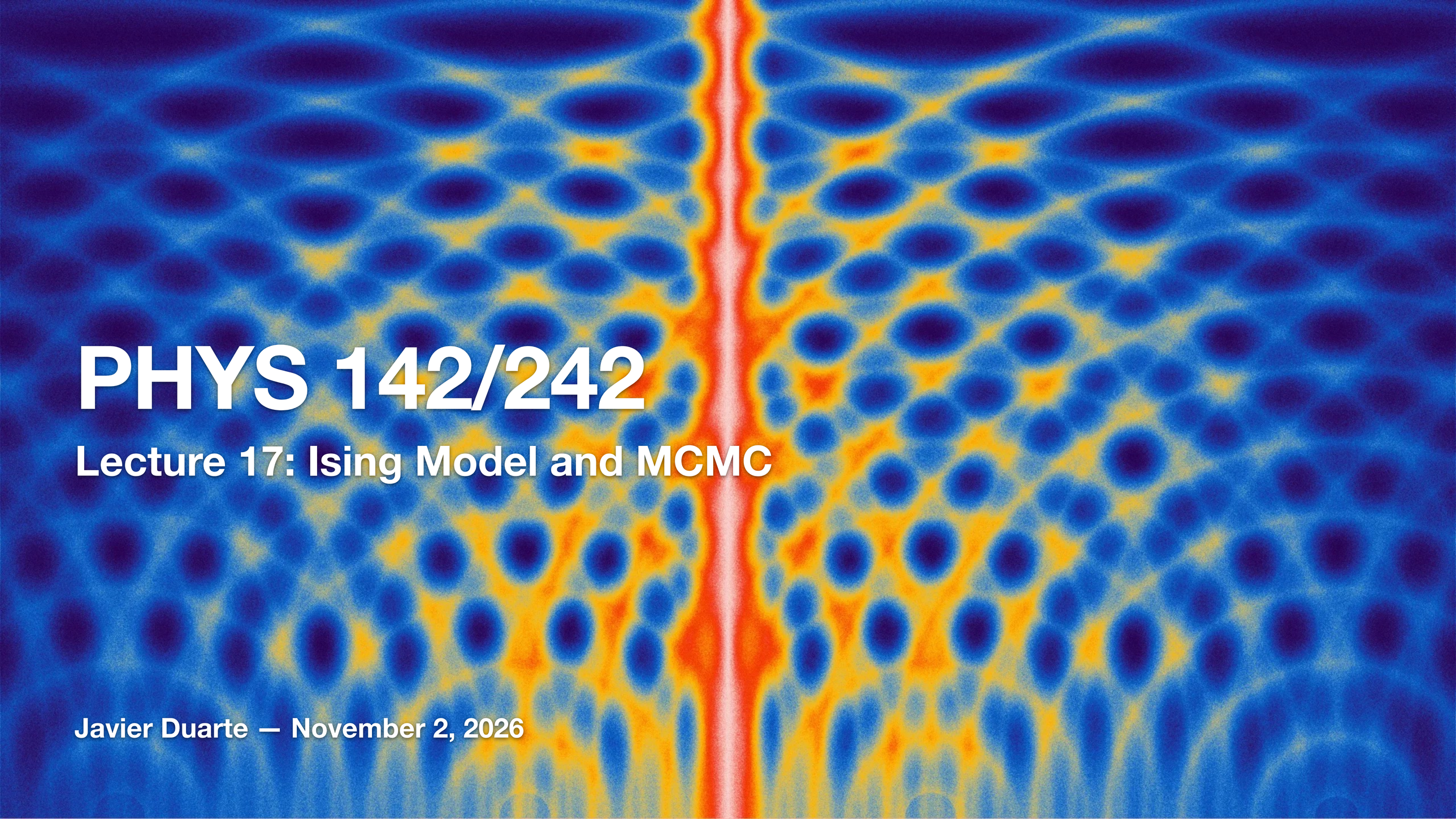


PHYS 142/242

Lecture 17: Ising Model and MCMC

Javier Duarte — November 2, 2026

2D Ising model

The Hamiltonian is

$$E(\sigma) = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j - h \sum_k \sigma_k$$

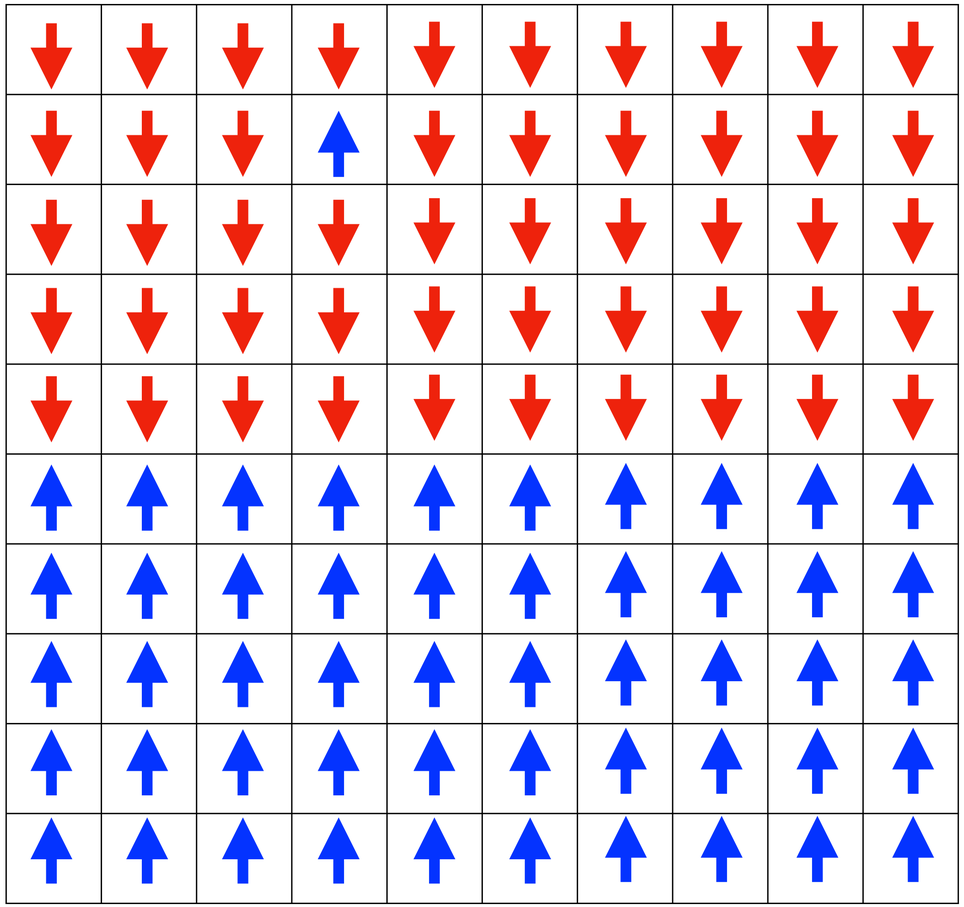
where $\sigma_k \in \{-1, +1\}$ represents the spin of each site. The probability of any given state is

$$p(\sigma) = \frac{1}{Z} \exp\left(-\frac{E(\sigma)}{k_\mathrm{B} T}\right)$$

where

$$Z = \sum_\sigma \exp\left(-\frac{E(\sigma)}{k_\mathrm{B} T}\right).$$

With 10×10 sites $2^{100} \approx 10^{30}$ possible states



Energy

Question: Given the Hamiltonian with no external field, what is the energy of the configuration here?

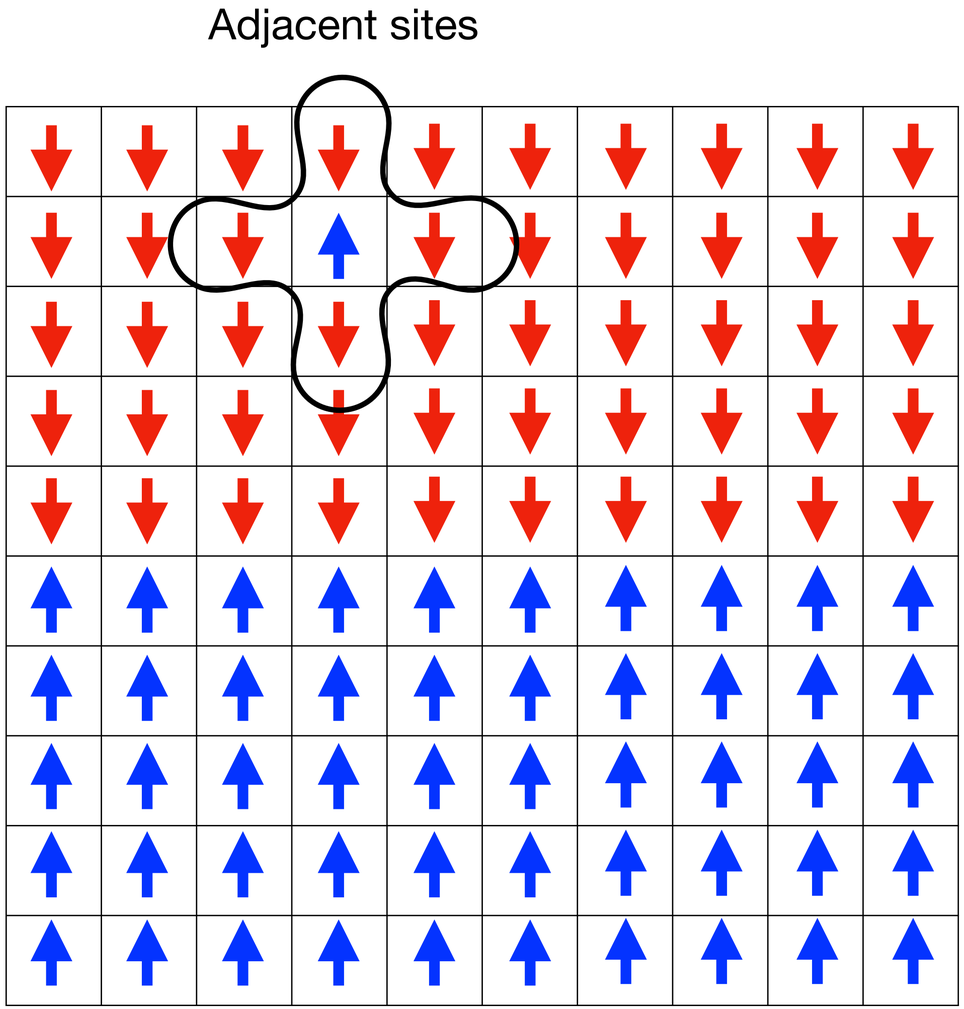
$$E(\sigma) = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

Assume periodic boundary conditions

Hint: number of adjacent site pairs (no double counting) is

$$N_{\text{adjacent pairs}} = 2\,|\,\Lambda\,|$$

Answer: The energy of the configuration here is $E = -152\,J$

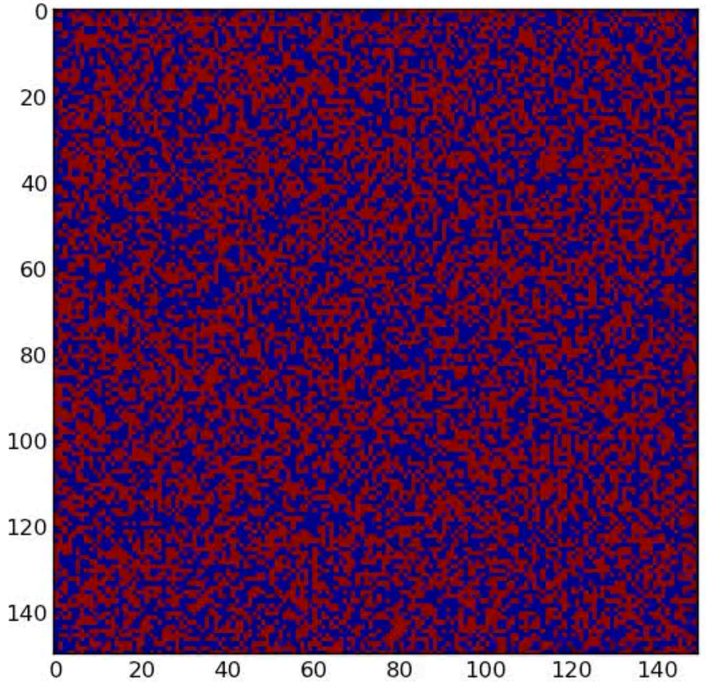


Ising model phase transition

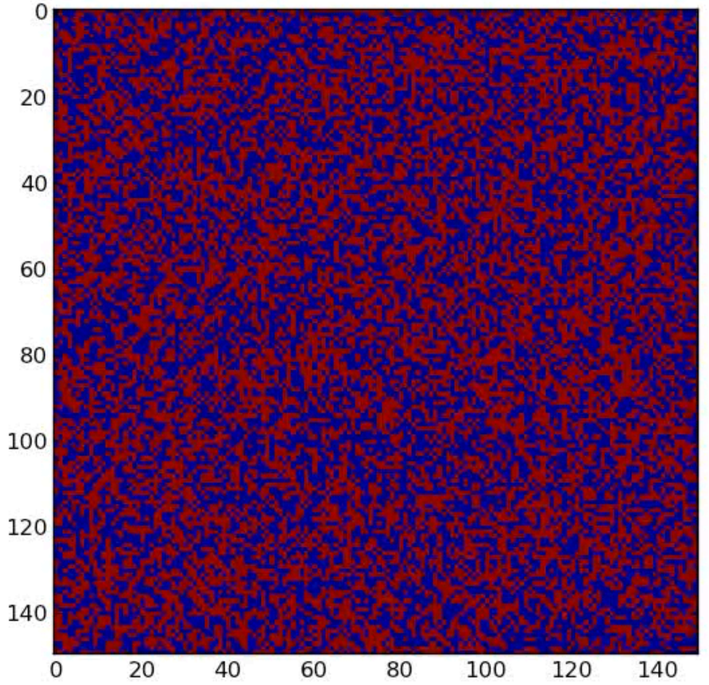
With 150×150 lattice, $2^{22500} \approx 10^{6773}$ possible states!

Phase transition occurs at a critical temperature: ordered phase at low T, disordered phase at high T. How can we compute this?

High T



Low T



Ising model: magnetization

Often interested in quantities like the average magnetization, i.e. first we average over the lattice

$$M(\sigma) = \sum_{k \in \Lambda} \sigma_k, \qquad m = \frac{M}{N} = \sigma_{\text{ave}}, \qquad N = |\Lambda| = L^2$$

then we take a thermodynamic average

$$\langle m \rangle = \sum_{\sigma} m(\sigma) p(\sigma)$$

If we plot $\langle m \rangle$ versus T, we can see there is some critical temperature T_c where the system goes from a ferromagnetic phase (spins are aligned) to a paramagnetic phase (spins are not aligned)

Ising model: specific heat

Also interested in the **specific heat** (energy required to raise the temperature of a given mass by one degree)

Recall $U = -\frac{\partial \ln Z}{\partial \beta} = -\frac{1}{Z}\frac{\partial Z}{\partial \beta} = \frac{1}{Z}\sum_n E_n e^{-\beta E_n} = \langle E \rangle$

$$C = \frac{\partial U}{\partial T} = -\frac{1}{k_B T^2}\frac{\partial U}{\partial \beta} = \frac{1}{k_B T^2}\frac{\partial^2 \ln Z}{\partial \beta^2}$$

$$C = -\frac{1}{k_B T^2}\left(\frac{1}{Z^2}\left(\sum_n E_n e^{-\beta E_n}\right)^2 - \frac{1}{Z}\sum_n E_n^2 e^{-\beta E_n}\right) = \frac{\langle E^2 \rangle - \langle E \rangle^2}{k_B T^2}$$

Takeaway: MCMC can give us the specific heat from energy fluctuations at a single temperature T

Ising model: susceptibility

With an external magnetic field h, can also consider the ***(magnetic) susceptibility***: how easily a material becomes magnetized in an external magnetic field

$$\chi = \left.\frac{\partial \langle M \rangle}{\partial h}\right|_{h=0} = \frac{\langle M^2 \rangle - \langle M \rangle^2}{k_B T}$$

Per-spin versions of the specific heat and susceptibility are conventional:

$$C/N = c = N(\langle u^2 \rangle - \langle u \rangle^2)/(k_B T^2)$$

$$\chi/N = N(\langle m^2 \rangle - \langle\, |m| \,\rangle^2)/(k_B T)$$

During phase transition, both peak at T_c, so highlight T_c better than m

Ising model: comparison to theory

In 2D, we can actually analytically calculate the critical temperature as solved in L. Onsager, Phys. Rev. 65, 117 (1944)

$$\frac{k_{\mathrm{B}}T_{\mathrm{c}}}{J} = \frac{2}{\ln(1+\sqrt{2})} \approx 2.269$$

and find behavior of magnetization as in N. Yang, Phys. Rev. 85, 808 (1952)

$$m_0(T) = \left[1 - \sinh^{-4}\left(\frac{2J}{k_B T}\right)\right]^{1/8} \quad (T < T_c), \qquad u(T_c) = -\sqrt{2}\,J$$

Critical exponents

Phase transitions are often characterized by how physical quantities behave around the critical temperature T_c usually as a power law (or logarithm)

For the 2D Ising model, near T_c, we have

$m \sim (T_c - T)^{1/8}$,

$\chi \sim |T - T_c|^{-7/4}$,

$\xi \sim |T - T_c|^{-1}$, and

c grows like a logarithm

So the ***critical exponents*** are $\beta = 1/8$, $\gamma = 7/4$, $\nu = 1$, and $\alpha = 0$ (log)

MCMC vs. mean-field theory

Why **simulate**? Other approaches actually get behavior like the critical temperature and exponents wrong!

For example, ***mean-field theory*** replaces the fluctuating interactions of neighboring spins with an effective average, turning a complex interacting lattice problem into a solvable single-spin problem

It predicts (self-consistency relation)

$$m = \tanh((4Jm + h)/k_B T)$$

This gives $\frac{k_B T_c}{J} = 4$ (76% too high), and $\beta = 1/2,\ \gamma = 1,\ \nu = 1/2,\ \alpha = 0$

Applying MCMC to the 2D Ising model

Assuming $h = 0$,

1. On step k, randomly choose one of the spins i and consider flipping it $\sigma_i \to -\sigma_i$
2. Calculate the change in energy that would result from flipping spin i, i.e. the quantity:

$$\Delta E = 2J\sigma_i \sum_{j \in \mathcal{N}(i)} \sigma_j$$

   If $\Delta E \leq 0$, accept the spin flip

   If $\Delta E > 0$, accept the spin flip with probability $\exp(-\Delta E / k_{\mathrm{B}} T)$. Otherwise, reject the flip.

3. Update the moving average of $\langle \sigma_{\mathrm{ave}} \rangle$ (or whatever quantity we are interested in).

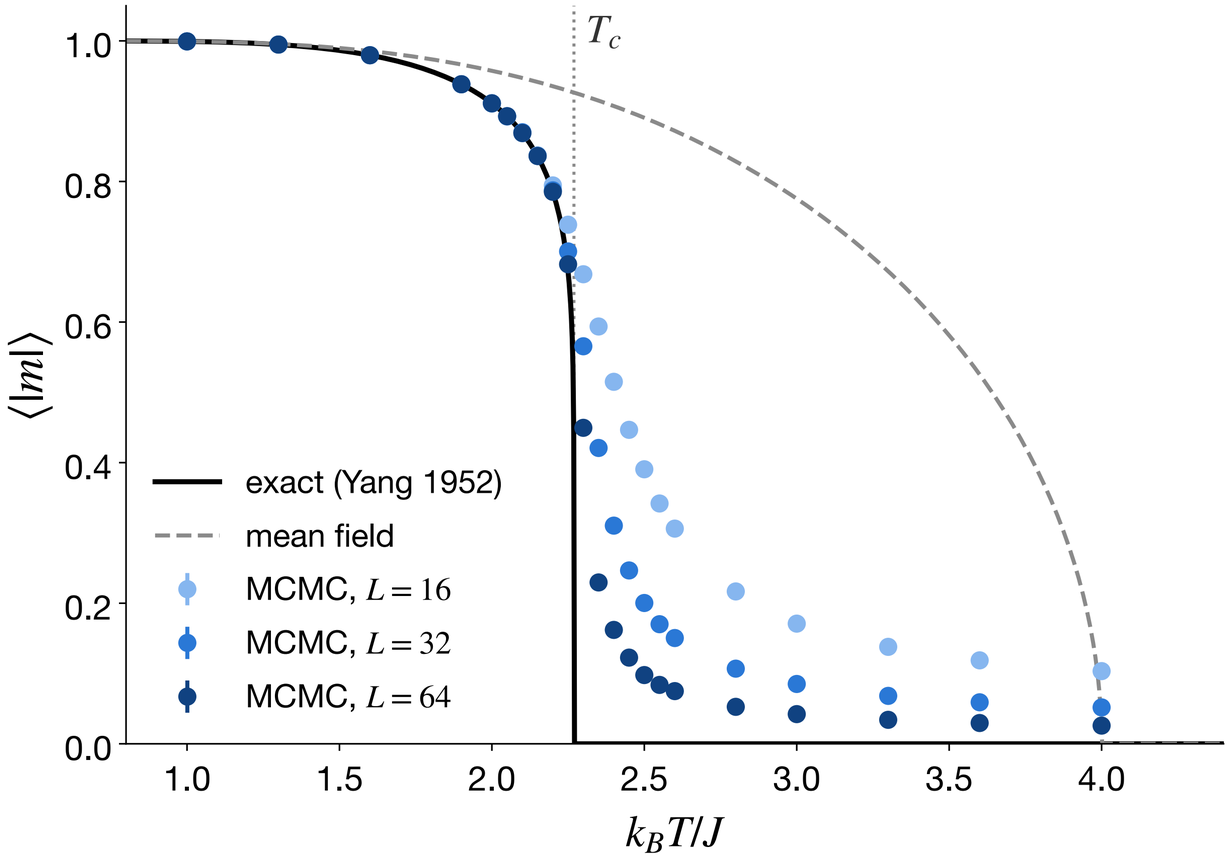
4. Repeat.

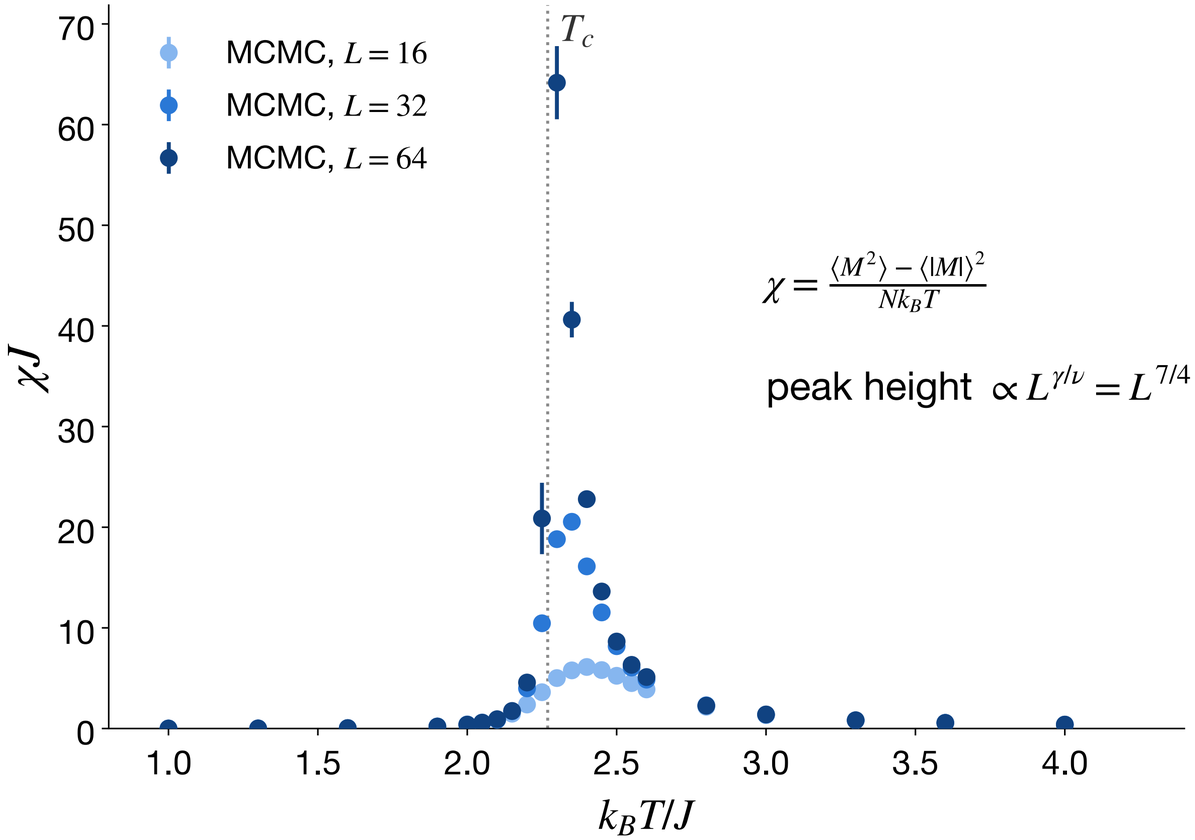
Practical considerations for Ising

- Several practical considerations
  - **Cleaner** ΔE: ΔE can only be -8, -4, 0, 4 or 8 (in units of J), so you can precompute the only two acceptance probabilities ever needed
  - **Count time in sweeps** (one sweep is N attempted flips). 1M steps on a 100×100 lattice is only 100 sweeps, with 10 sweeps of burn-in.
  - **Acceptance depends on** T: single-spin flips have no step size to tune, so the temperature sets the acceptance rate
  - **Critical slowing down**: at T_c the integrated autocorrelation time grows as L^z, with $z \approx 2.17$ for Metropolis. More sophisticated "cluster" algorithms (Wolff, Swendsen-Wang) can get improved $z \lesssim 0.3$

Results: magnetization and susceptibility

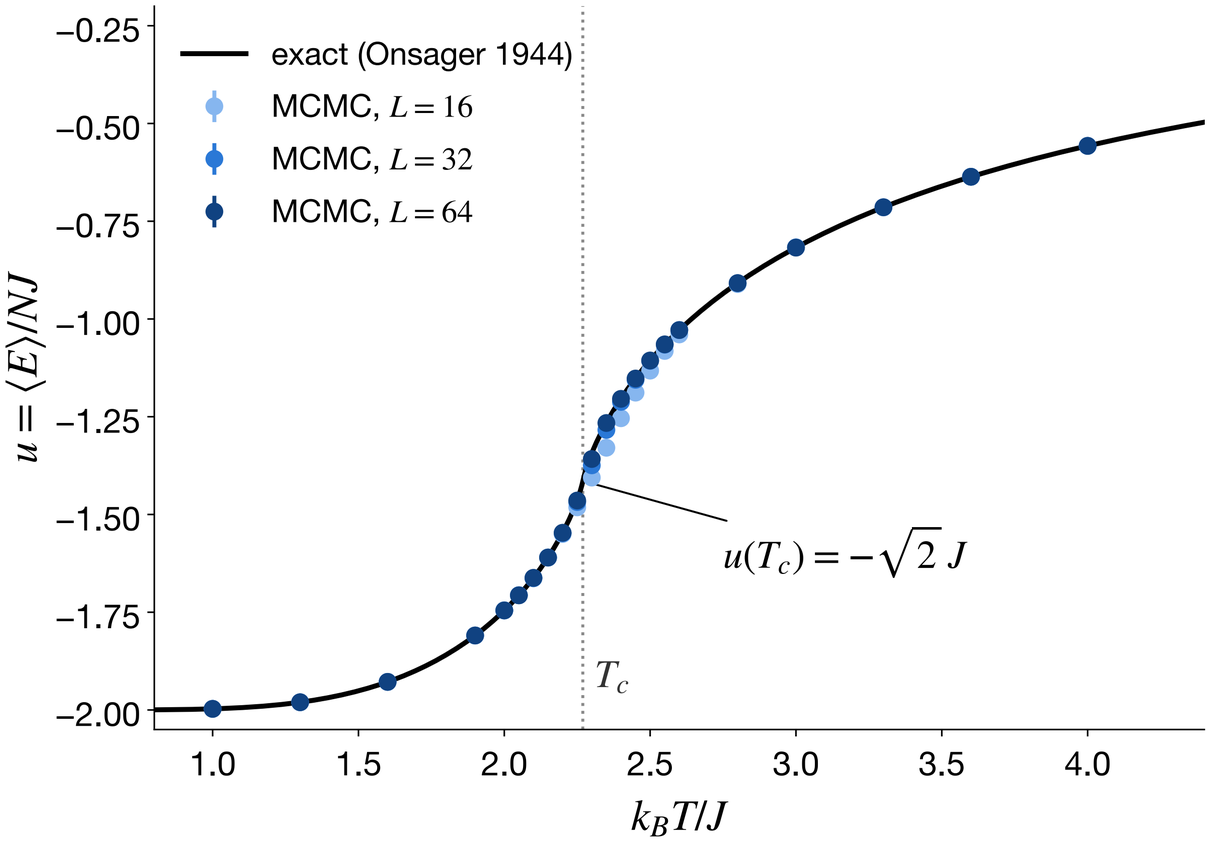
- Magnetization and susceptibility
- Note: the peak height grows $\propto L^{7/4}$

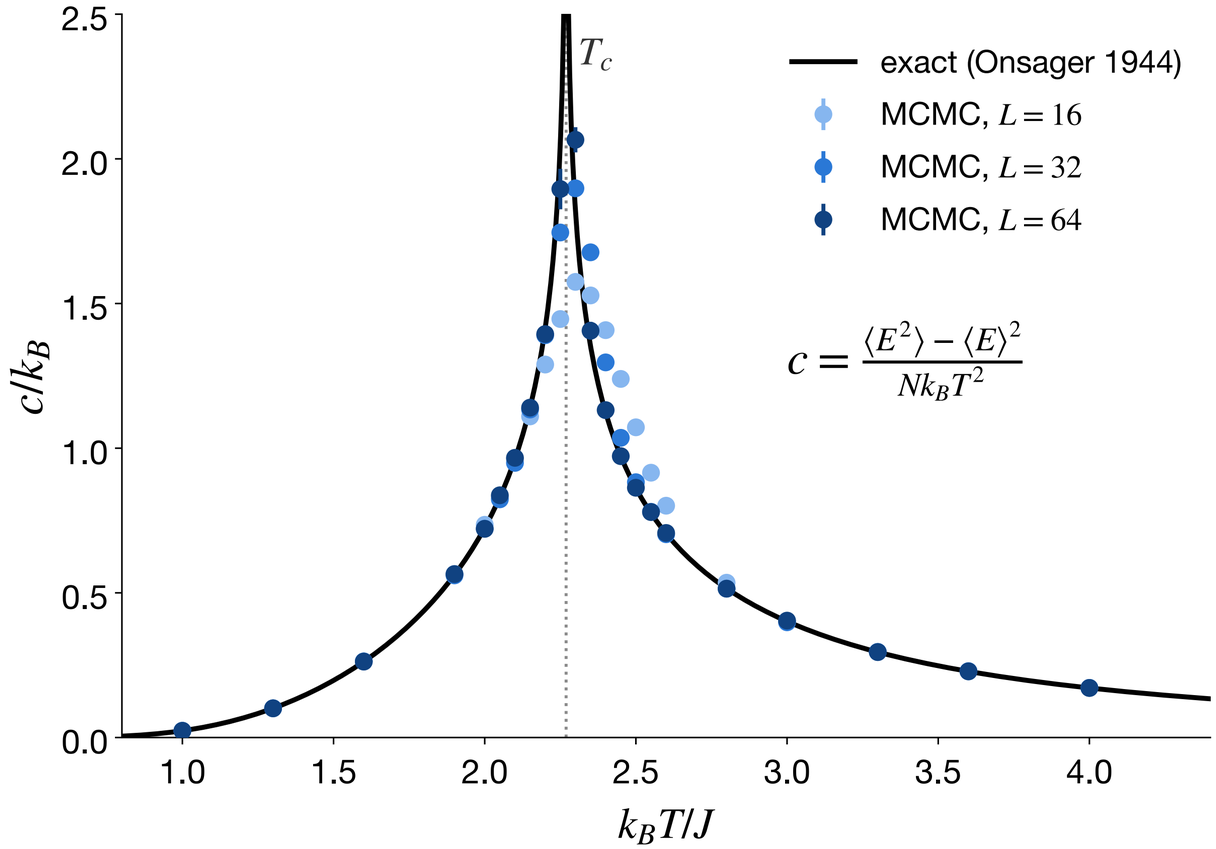




Results: energy and specific heat

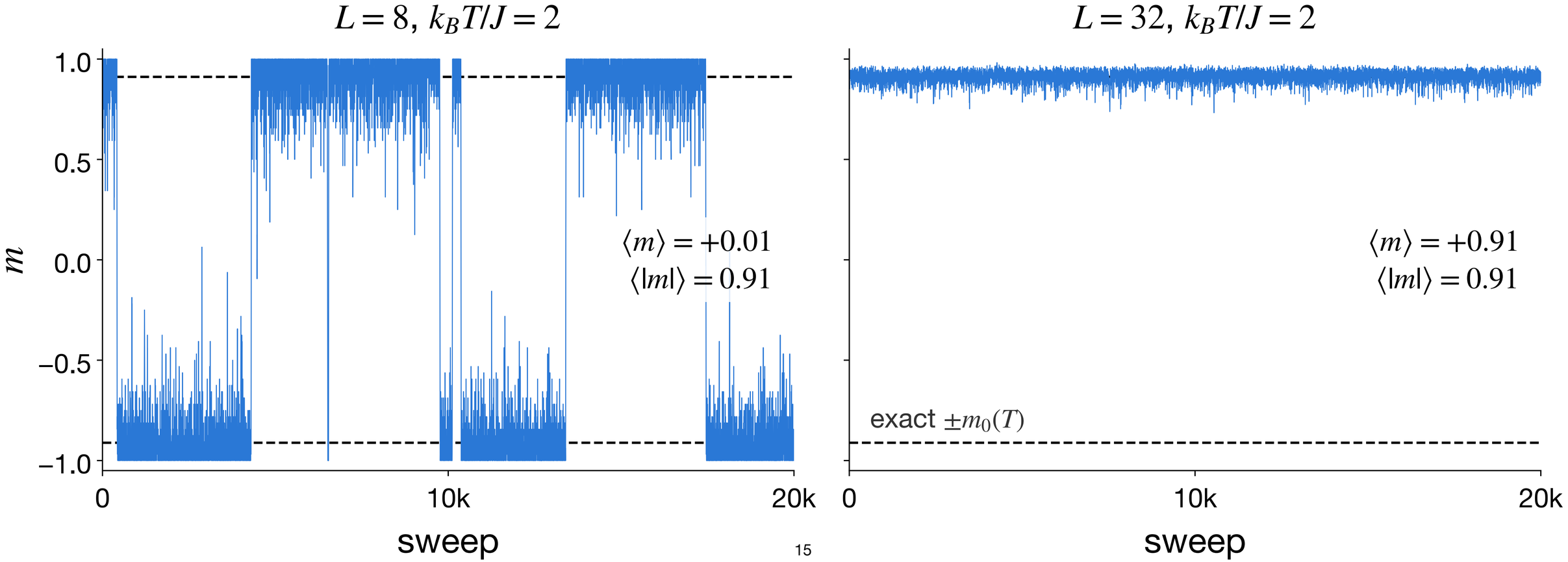
- Energy and specific heat





Pitfall: $\langle m \rangle = 0$ by symmetry

- $E(\sigma) = E(-\sigma)$, so $\langle m \rangle$ is exactly zero on any finite lattice; use $\langle\,|\,m\,|\,\rangle$
- Below T_c, small lattices tunnel between $+m$ and $-m$ and large ones never do (similar to double well problem)



Critical slowing down

- Autocorrelation time peaks at T_c and grows $\propto L^z$ (fit $z \approx 2.0$; literature 2.17)
- Wolff and Swendsen-Wang cluster updates can improve ($z \lesssim 0.3$)

